

Semi-Autonomous Robotic system for efficient recycling of Lithium-Ion Batteries

Shreyash Gadgil ^{1,†} , Durva Gaikwad ^{1,†}, Bijo Sebastian ^{1,*} , Alpana Dubey ² and Abhinav Upadhyay ²

¹ Department of Engineering Design, IIT Madras

² Accenture Labs, Accenture

* Correspondence: bijo.sebastian@iitm.ac.in

† These authors contributed equally to this work.

Abstract: The increasing popularity of electric vehicles is leading to a massive increase in the use of lithium-ion batteries (LIBs), majority of which will reach their end-of-life in the next decade. Current recycling methods employed to recover valuable materials from end-of-life battery systems involve shredding of individual cells, followed by the extraction of materials through chemical or metallurgical processes. However, this approach often leads to a reduction in the overall yield of valuable materials, and it results in the wastage of other auxiliary components. The expected environmental impact of end-of-life LIBs necessitates sustainable recycling solutions. This paper introduces a semi-autonomous robotic system for efficient recycling of LIBs, showcasing a successful collaboration between human, robot, and a disassembly mechanism. The proposed system leverages human skills in removing fragile and deformable battery casing, intelligent grasping capabilities of a robotic manipulator in retrieving individual cells from the battery and the reliability of a disassembly mechanism in extracting the electrode stack from individual cells. By eliminating risks associated with manual recycling and the wastage associated with conventional shredding, the proposed concept allows for direct recycling and reuse of LIB components. While detailed experimental validation of the proposed semi- autonomous robotic system is beyond the scope of this concept paper, the idea along with the proof-of-concept validation presented here, could drive the discussion towards further research and large scale adoption of semi-autonomous or autonomous robotic systems for LIB waste disposal, and thereby reduce the environmental impact from end-of-life batteries.

Keywords: lithium-ion batteries, robotics, recycling, circular economy, semi-autonomous disassembly.

Citation: Gadgil, S.; Gaikwad, D.; Sebastian, B.; Dubey, A.; Upadhyay, A. Semi-Autonomous Robotic system for efficient recycling of Lithium-Ion Batteries. *Appl. Syst. Innov.* **2023**, *1*, 0. <https://doi.org/>

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors. Submitted to *Appl. Syst. Innov.* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Climate change poses a significant threat to both our environment and humanity, with transportation being a major contributor to greenhouse gas emissions[1,2]. To mitigate this, numerous countries globally are advocating for the adoption of electric vehicles (EVs). Unlike traditional vehicles that rely on complex internal combustion engines, EVs operate on an electric drive train. However, these vehicles require substantial battery systems to store energy. With the size and weight of these batteries directly affecting the vehicle's range, a key factor in their competitiveness, vehicle manufacturers are trying to include larger, higher capacity battery packs in their upcoming EV models.

The preferred choice for energy storage in these vehicles is often Lithium-ion batteries (LIBs). Their dominance is attributed to the high energy density, both volumetric and gravimetric, their longevity, low maintenance requirements, and high-current capabilities. These features enable LIBs to satisfy user demands for EVs with high speed, torque, and range [3].

In the coming decades, the demand for EVs is expected to further increase; resulting in exponential growth for the lithium-ion market [2]. The demand for LIBs is projected to surge, with an estimated annual growth rate of about 27 percent. A projected 23 million

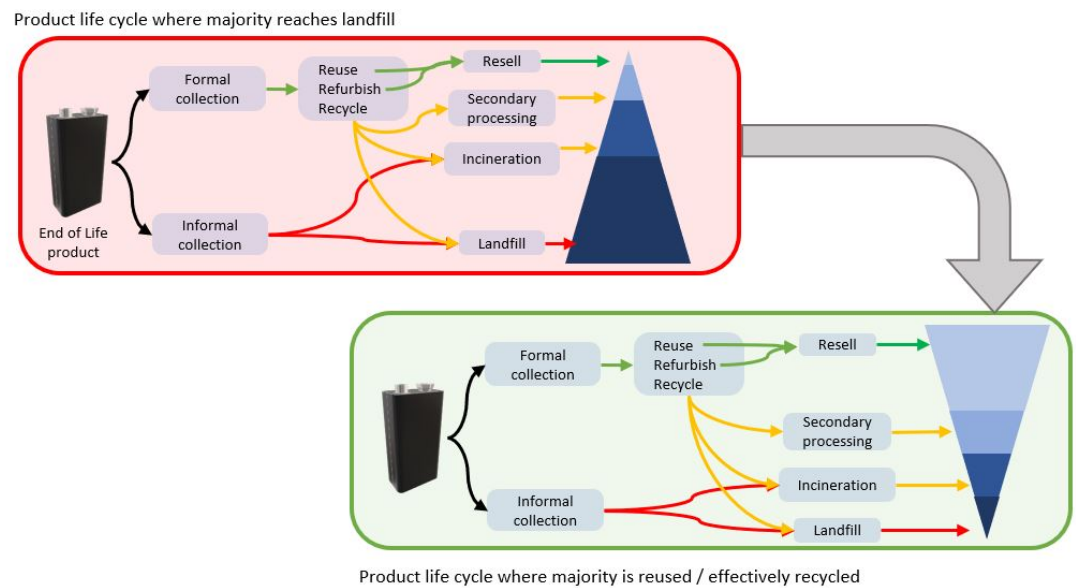


Figure 1. Achieving circularity by inverting the end-of-life pyramid

EV cars sold globally in 2030 might result in 5,750,000 tonnes of retired batteries by 2040, assuming a battery life of 10 years and a battery pack weight of 250 kg [4]. This growing demand for LIBs brings up the substantial concern regarding increasing number of end-of-life lithium-ion batteries (LIBs) as the production and recycling rates of lithium-ion batteries (LIBs) differ significantly[8]. Unfortunately, only a small percentage of LIBs are recycled at the end of their lives, leading to a product life cycle where, the vast majority ends up in landfills, as shown in Figure 1. Addressing the current gap between recycling and production of LIBs is important from an economic and environmental standpoint.

1.1. Recycling of LIBs: Economic impact

In recent years, circular economy has gained pace as a greener solution to raw material production. The core principles of a circular economic approach are: (1) eliminate waste and pollution, (2) use products and materials for a more extended period, and (3) re-generate natural ecosystems. Recycling facilitates internal material movements and plays a critical role in the circular economy (CE) [12–15].

Applied in the context of LIBs, recycling and reuse allows for the constituent materials to be used for a longer period and act as a viable option for restoring LIB chemicals into the economic cycle and minimizing reliance on primary raw materials [9–15]. As shown in Figure 1, efficient recycling techniques would allow us to invert the end-of-life pyramid for LIBs, leading to better circularity in the use of raw materials. We can not only reduce economic and environmental problems by closing the loop through proper recycling, but we can also create a more sustainable and resource-efficient approach to LIB production and consumption.

1.2. Recycling of LIBs: Environmental impact

Existing recycling techniques largely focus on removing economically useful minerals with direct economic value such as gold, copper, iron, and aluminium, resulting in the active material, especially Lithium oxide, left behind as waste. Complete disposal of LIBs or the active materials present in LIBs in the form of landfills would impose an enormous threat to the natural environment and human health[17–19]. Fire and explosion incidents during end-of-life disposal of batteries, as well as leaching after disposal, could create future pollution pathways. Hazardous substances released from LIBs include vapours, gases, metal nano oxides, and degradation products of the electrolyte. These pollutants can

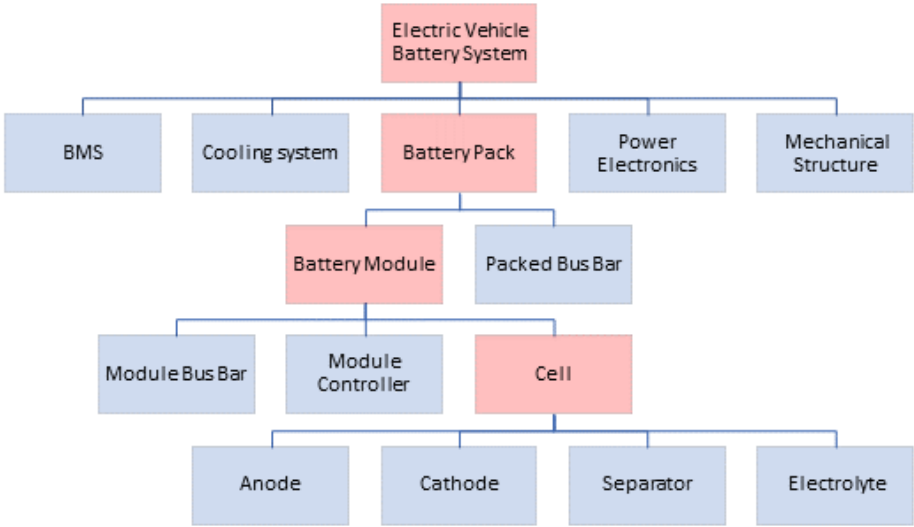


Figure 2. Electric vehicle battery structure.

be released into soil, water, and air, posing a direct threat to animal and plant species at all trophic levels, as well as a risk to human health. [19].

In addition to addressing the issues of LIB disposal, it is critical to acknowledge the enormous environmental impact of mineral extraction and smelting operations. These processes can exacerbate soil erosion and dust phenomenon, aggravating environmental concerns.

The above environmental concerns will make effective large-scale recycling of LIBs a necessity in the coming decades. To address the same, this paper presents a semi-autonomous approach towards efficient recycling of LIBs.

The rest of the paper is organised as follows; Section 2 provides an overview of LIB design and existing autonomous, semi-autonomous and manual LIB recycling techniques. Section 3 presents the overall layout of the Semi-Autonomous disassembly. Section 4 discusses the experimental validation using the proposed system on a mock battery. Section 5 concludes the paper with suggestions for future research directions.

2. Literature Review

The following section provides a review of the commonly used LIB designs. The section also provides a review of existing disassembly techniques for LIBs aimed at recycling and reuse of battery material. This review will motivate the need for the proposed semi-autonomous robotic disassembly of LIBs.

2.1. Lithium-ion batteries

Lithium-ion batteries, particularly the type used in electric vehicles (EVs), are structured hierarchically in a "cell-module-pack" configuration, as shown in Figure 2. Because of the high-power requirements of EVs, a large number of battery cells, ranging from dozens to thousands, are required. These cells serve as the base of the battery and must have a high capacity per unit volume in order to optimize performance within the confines of a vehicle’s available area. Furthermore, they must have long cycle life and be able to survive the mechanical shocks, vibration, and temperature fluctuations encountered while driving.

To meet the above performance requirements, multiple cells are built into modules within protective frames. The final battery pack will consist of multiple modules, a Battery Management System (BMS), and a cooling device, in addition to the power electronics com-

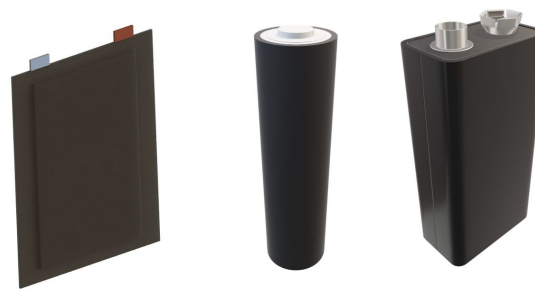


Figure 3. Typical cell types used in lithium-ion batteries: Pouch cell, Cylindrical cell, and Prismatic cell.

ponents and the mechanical structure itself. The BMS is responsible for voltage and current regulation during charging and discharging cycles. The cooling system is responsible for maintaining the temperature of the overall system within limits during operation.

Each individual cell consists of four essential components, namely cathode, anode, electrolyte, and separator. The active material, conductive additive, and binder are all contained within the cathode, which is deposited on a thin aluminium foil. The active material, lithium-ion in this case, determines the capacity of the battery. The conductive additive improves conductivity, and the binder works as an adhesive. The anode, also coated with active material, allows electric current to flow and the reversible absorption/emission of lithium ions. The electrolyte enables the movement of lithium ions between the cathode and anode, ensuring operation of the battery while maintaining safety.

Lithium-ion cells, comes in three primary form factors: cylindrical, prismatic, and pouch cells[20], as shown in Figure 3. Each of these form factors has its unique advantages and disadvantages, and their selection is primarily driven by specific application requirements.

Among the above designs, currently the pouch and the prismatic cells have higher growth rates in the global market [21]. These battery types offer superior heat management capabilities compared to cylindrical batteries, which can impact overall performance. Pouch cells employ a unique approach where the cells are sealed in a flexible foil container instead of a solid enclosure. Recent pouch cell designs employ a z-folded stacking, as shown in Figure 7 (a). The Z-stack structure consists of alternating layers of positive and negative electrodes, separated by a separator material. The flexible cover with the z-folded stacking technique allows for improved energy density and better space utilization within battery modules. This design reduces weight and allows for flexible cells that can conform to the available space, resulting in high energy density, making them highly desirable for a wide range of applications, including portable electronics and electric vehicles. However, these cells can swell and rupture when subjected to mechanical or thermal shocks, leading to safety concerns and need for better packaging at the battery level.

Existing literature consistently highlights the potential of pouch cell batteries for future applications, emphasizing their versatility, energy efficiency, and compatibility with emerging technologies. As such, the proposed disassembly approach would be designed for and validated on pouch cell batteries with z-folded stacking.

2.2. Review of existing recycling practices

Existing efforts towards recycling and reuse of LIBs primarily involve manual disassembly of the cells and electrode stack, followed by electrochemical treatment of the electrode material to extract electrode metals and the active material, lithium oxide. Among the existing LIB recycling strategies, direct manual disassembly and recycling is the most widely used, especially in developing nations. It involves manual removal of cathode or anode material from the electrode for reconditioning and re-use in a remanufactured LIB.

This method has been primarily applied to laptop and mobile phone batteries due to their abundance[24].

Direct manual disassembly involves high risk. Residual charge left in the battery or individual cell could result in electrical shocks, leakage of harmful chemicals, explosion and fire, during the disassembly process. As such, specialized equipment is required for the manual disassembly to prevent safety hazards. Ceramic blades are recommended for cutting open of individual cells to prevent accidental short circuit. It is also recommended to perform manual disassembly while keeping the individual cells inside a glove chamber that maintains an inert atmosphere. This will help prevent accidental fire or poisonous chemical leakage during the process. The pouch and separator materials used in the battery cells have lower economic worth when compared to other battery cell components. As such, traditional manual recycling processes do not prioritize the retrieval of these materials [16]. Manual disassembly approaches are also not scalable for handling large volumes of end-of-life LIBs, anticipated at 300 million tonnes from electric vehicles alone by the year 2040[23]. All of the above factors make manual disassembly of LIBs expensive, dangerous and wasteful.

In addition to direct manual disassembling, other major methods used to recycle lithium-ion batteries are pyrometallurgy, and hydrometallurgical treatment. Pyrometallurgy involves melting batteries in a high-temperature furnace to convert metal oxides into a Co, Cu, Fe, and Ni alloy[24]. This method is adaptable and can handle unsorted feedstock and batteries mixed with other waste. It produces a metallic alloy fraction, slag, and gases. Pyrometallurgy while preferable due to its simplicity and efficient processing, does not allow for recovery of electrolytes, plastics, or other components. Pyrometallurgical process also leaves a larger negative impact on environment.

Hydrometallurgical operations typically start with battery shredding and black mass separation, the solid residue that remains after the extraction of valuable metals from spent lithium-ion batteries. Aqueous solutions are used to leach desired metals from cathode materials in LIBs. The metals are extracted as sulphate, oxalate, hydroxide, or carbonate, while lithium can be precipitated [24]. Shredding reduces the size and passivates reactive components but yields lower purity products, affecting process economics. While shredding results in size reduction and passivation of reactive components, it also yields lower purity products and thereby decreasing process economics [22]. It has been shown that the cost saving using shredded material was generally less than 20%, whereas disassembly could potentially result in cost savings in the range of 40% to 80% depending on purity[22].

Among the three recycling strategies, direct manual recycling stands out for its process simplicity, minimal environmental impact, and profitability among the three [25]. But it suffers from the disadvantages and limitations associated with manual disassembly, as mentioned earlier.

3. Design of Semi-Autonomous Disassembly System

Based on the detailed literature review, there is an immediate need for the development of a scalable, efficient, and economical approach towards LIB disassembly and recycling. As mentioned in the review section, a majority of the existing efforts towards recycling and reuse of LIBs involve manual disassembly. The manual aspect results in high cost, primarily driven by labour expenses and the safety considerations necessary to prevent electrical and chemical hazards during the manual disassembly. Additionally, many of the disassembly steps involved in battery recycling are repetitive and time-consuming. The above challenges make LIB recycling an excellent case for automation, which would also make it economically viable and scalable to handle the increasing demand for batteries.

Automation has revolutionized industries especially in performing quick, precise, and repetitive tasks, especially in hazardous environments where human safety is a concern. However, it is important to acknowledge that while automation excels in some aspects, it lacks the inherent flexibility, and adaptability possessed by humans. Traditional factory robotics and automation rely on highly organised environments in which robots execute

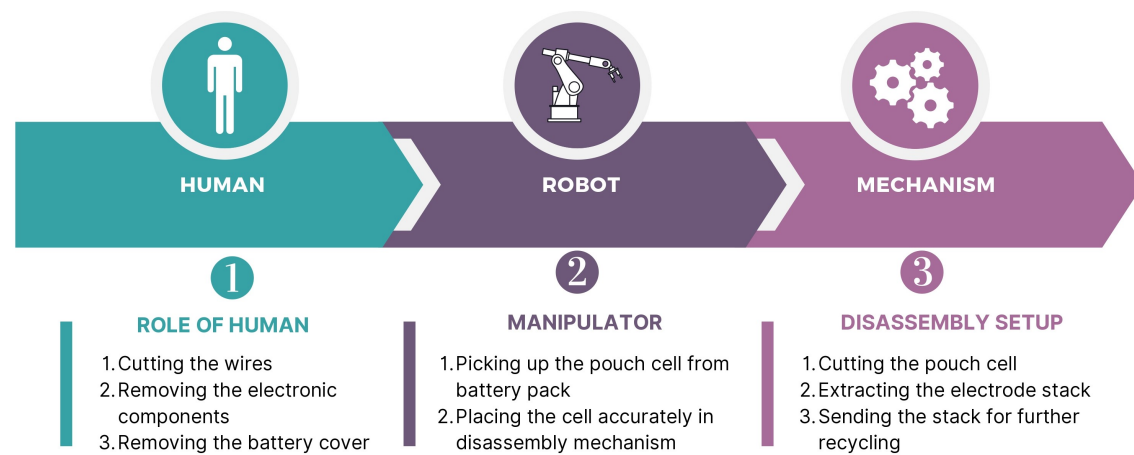


Figure 4. Semi-autonomous battery disassembly process flow

repeated pre-programmed operations on precisely known items in predetermined configurations. In contrast, the presence of soft, fragile, and flexible components in batteries such as soldered wires, adhesive seals, and insulating covers makes it difficult to develop a fully automated system for addressing the disassembly requirements. Moreover, lack of standardisation in battery pack, module, or cell design in the automotive industry also introduces variations in the disassembly process, making it difficult to address through a fully automated system [28].

Wegener et al. [26] have recommended a hybrid approach involving both human and automation for EV battery disassembly. By combining the strengths of both automation and human workers, a semi-autonomous battery disassembly system could be created, that maximizes efficiency, productivity, and safety. Automation can handle repetitive and mundane tasks, freeing up human workers to focus on more complex tasks. Humans can provide the necessary oversight, decision-making, and problem-solving skills that automation currently lacks. This would result in a scalable approach towards battery recycling while minimizing the safety risk to human employees. While the installation cost of such a system could be high, the running cost over large volumes would be lower, making recycling economically viable [26,27].

Keeping the requirements and inferences from existing literature in mind, we propose a semi-autonomous robotic approach towards the disassembly of LIBs. The proposed approach involves a human operator, a robotic arm, hereafter referred to as a manipulator, and a disassembly mechanism, working together to extract an electrode stack from an LIB pack. The proposed workflow is summarized in Figure 4. The human operator is required to remove the flexible components, while addressing the variations in the battery design. The robotic system would be used for performing the repetitive tasks, such as removing individual cells from the module. A Specifically designed disassembly mechanism will be used for the dangerous tasks associated with the disassembly, such as cutting open the individual cells to remove the electrode stack. The proposed approach is customizable for different battery designs and scalable to handle the increasing volume of LIBs, while ensuring safety of the human worker involved in the process. The following subsections will discuss the semi-autonomous battery disassembly process in detail, as well as present the design of an equivalent battery on which the proposed approach would be validated.

3.1. Semi-Autonomous battery disassembly process

The disassembling procedure for battery packs can be initially classified into two broad categories, as disassembly of battery pack and disassembly of individual cells. Despite changes owing to design features, the fundamental steps involved in the disassembly of an EV battery system, usually consists of similar steps[29–33], listed as:

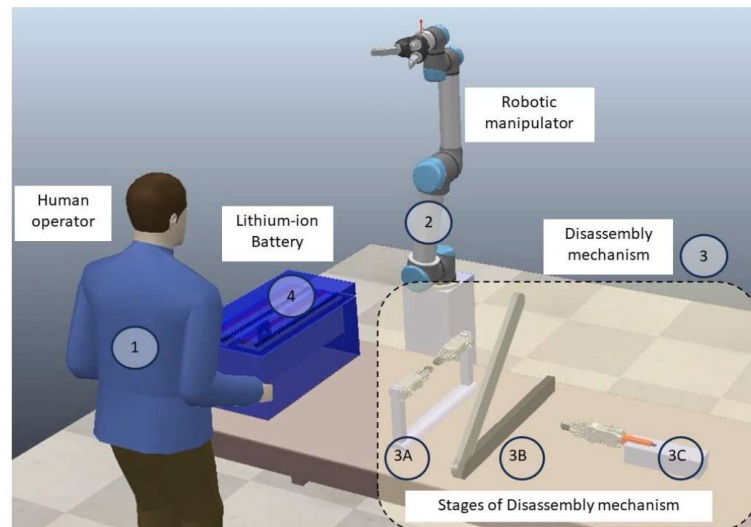


Figure 5. Overall layout of semi-autonomous disassembly system, showing the four sub-entities involved: human operator (1), robotic manipulator (2), a disassembly mechanism (3), and the LIB to be disassembled (4)

- a) Accessing the internal components of the battery system, which involves removing the protective cover 227
- b) Severing the electrical connections between the battery modules and the electronic components to isolate the individual parts 228
- c) Detaching the mechanical connections that secure the system components, such as the modules and electronics, to the battery base 229
- d) Removing the electronic components from the battery system 230
- e) Removing the battery modules from the system 231
- f) Dismantling the battery modules to extract the individual battery cells 232
- g) Dismantling the individual cells to extract anode, cathode and separator with the active material 233

Taking these steps into consideration, the overall layout of the semi-autonomous disassembly system is presented in Figure 5 and the steps involved in the disassembly process are detailed in Figure 6. The human operator is required to perform operations that require dexterity and cannot be accomplished by the robotic manipulator. This involves removing the top cover of the LIB, unscrewing and/or cutting open any seals on the battery as well as de-soldering any wires that secure the individual cells to the battery management system and the output terminals, as shown in Figure 6 a) and b). 234

The robotic manipulator and the disassembly mechanism together perform operations that are hazardous for the human operator. Repeated tasks such as picking up the individual pouch cells from within the LIB and accurately delivering it to the disassembly mechanism will be carried out by the robotic manipulator, as shown in Figure 6 c), d) and e). Further processing of individual cells to remove the electrode stack for recycling will be handled by the dedicated disassembly mechanism. The mechanism is divided into three distinct stages. The first stage focusing on the accurate positioning and gripping of the pouch cell, as shown in Figure 6 f). The second stage focuses on cutting open the pouch cell to expose the electrode stack, as shown in Figure 6 g). The third stage allows for retrieving the electrode stack from each pouch cell for further electrochemical processing, as shown in Figure 6 h). 235

3.2. Design of Equivalent Mock Battery 256

The wide variety of requirements for LIBs, even within the EV domain, results in the need for customization in terms of size, capacity, voltage, and other specifications. 257

As mentioned before, this leads to lack of standardization in LIB pack, module, or cell design, making it difficult to choose a particular battery model for validating the proposed semi-autonomous disassembly process. Lithium batteries contain toxic chemicals such as lithium, cobalt, and nickel, which can cause pollution and pose a risk to human health if not handled properly. As such, testing and validating a novel disassembly approach on a real LIB pack poses health and safety risk to the researchers involved.

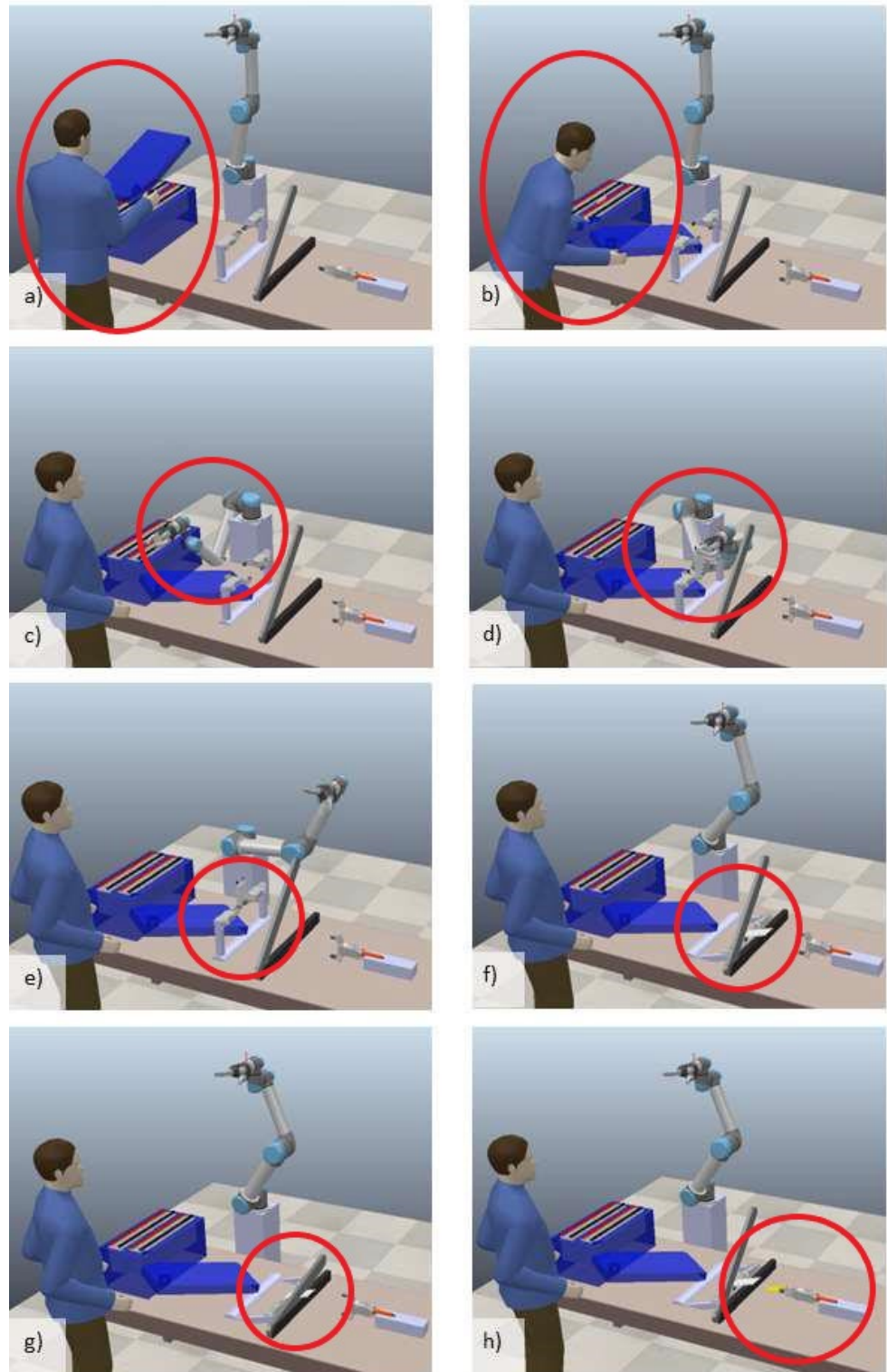


Figure 6. Steps involved in semi-autonomous LIB disassembly

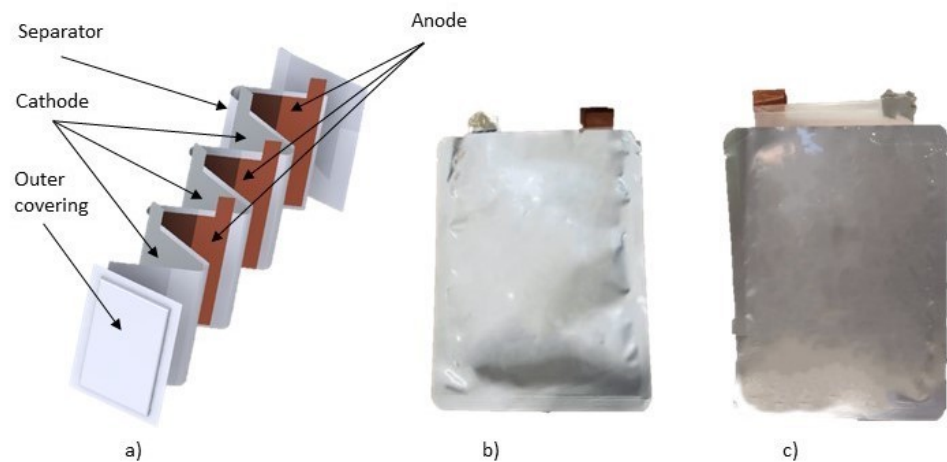


Figure 7. a) Exploded view of equivalent pouch cell, b) Equivalent pouch cell prototype, c) Equivalent pouch cell prototype with inner stack exposed

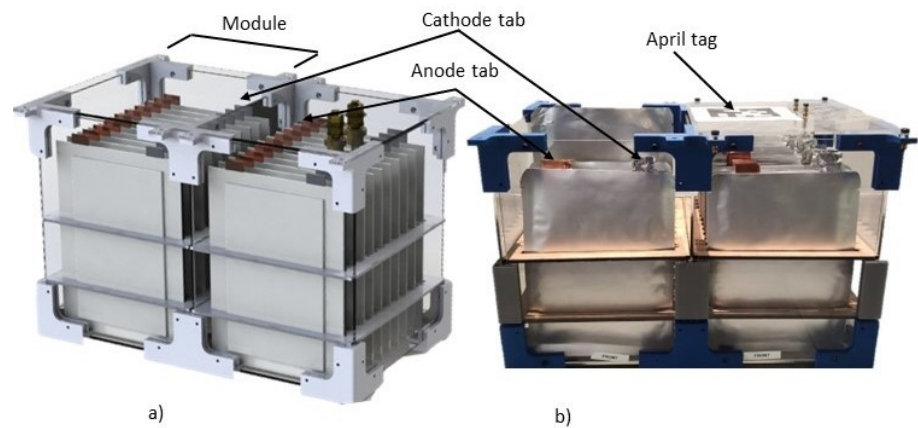


Figure 8. a) Design of equivalent battery, b) Physical model of equivalent battery

Taking the above facts into consideration, an equivalent LIB model was designed and fabricated to test, tune, and validate the proposed semi-autonomous disassembly technique. The pouch cells in the equivalent battery were made using materials that closely mimic the properties of actual pouch cells, while excluding the active materials like lithium-ion, due to its poisonous nature. Aluminium foil was used as cathode and copper foil as anode. Porous polypropylene sheets were used to replicate separator material. The outer cover of an LIB pouch cell typically consists of a composite sheet made of polymer and aluminium. A similar composite pouch of the same form factor was used in the equivalent battery. Similar to the LIB cells, a z-stacking design was used to pack the anode, cathode and separator into the pouch cell. The detailed CAD design of the equivalent pouch cell is shown in Figure 7 a), and the prototype is shown in Figure 7 b) and c). Ten pouch cells were stacked together to create a module. Finally, two modules were put together to create the equivalent battery pack.

The exterior of the battery was built using acrylic sheets held in place with 3D printed corner pieces. Figure 8 a) shows the CAD model annotating different parts of the equivalent battery. Figure 8 b) shows the integrated prototype.

4. Experimental Validation

This section presents the experimental validation of the proposed semi-autonomous disassembly approach on an equivalent battery. A detailed assessment of the proposed approach for large scale LIB recycling is also presented.

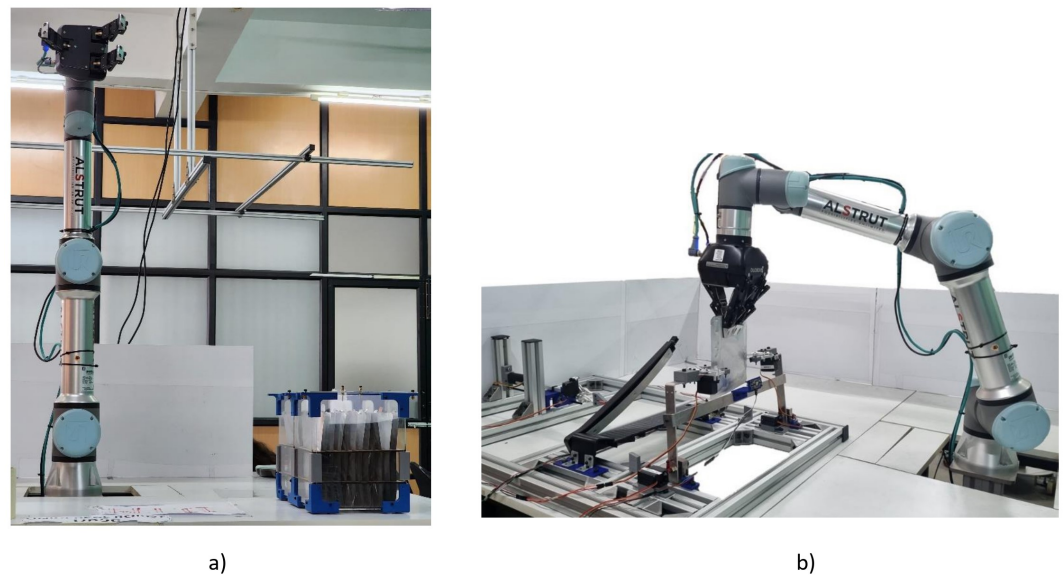


Figure 9. a)UR5e with Battery, b) UR5e with the disassembly setup

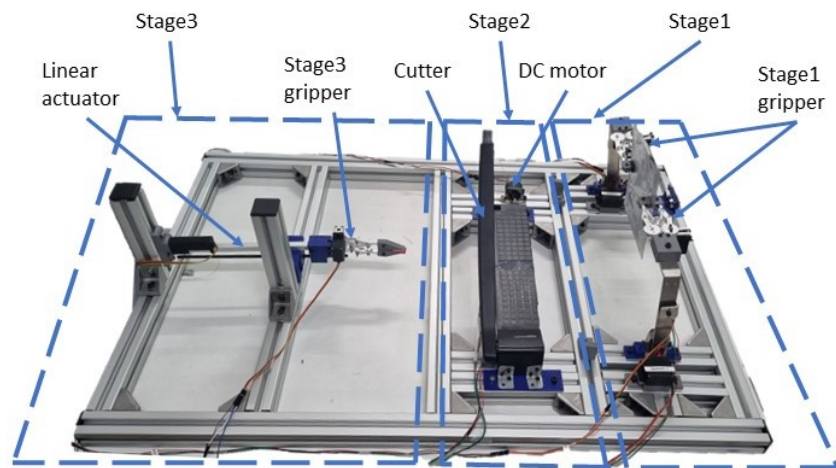


Figure 10. Layout of the disassembly mechanism

4.1. Experimental setup

Figure 9, and Figure 10 shows the experimental setup consisting of the equivalent battery, UR5e robotic manipulator and the disassembly mechanism. Figure 11 shows the steps involved in the disassembly process. To begin the disassembly process, a human worker is required to remove the top cover from the battery. The equivalent battery was provided with an AprilTag [5]. A calibrated camera was mounted overhead, which allowed for estimating the position and orientation of the equivalent battery and thereby the position of the pouch cells within the battery. This information was used to plan a feasible path for the UR5e manipulator from its starting configuration as shown in Figure 9 a) to the goal configuration as shown in Figure 11 a), where it can precisely grasp the pouch cell from the battery module. Autonomous path planning with obstacle avoidance and grasp planning was implemented for the UR5e manipulator using MoveIt [6] and ROS[7].

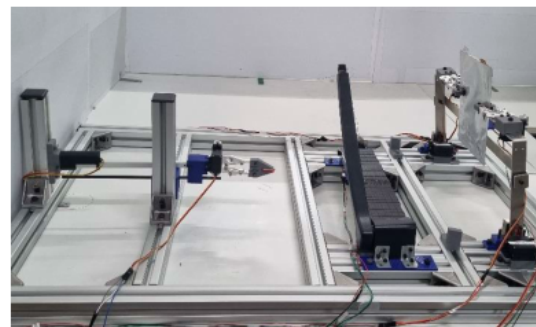
After successfully grasping the pouch cell, the MoveIt setup is used to autonomously plan a feasible path for the UR5e manipulator with the starting configuration as shown in 11 a) to the goal configuration as shown in 11 b). At this goal configuration, the UR5e manipulator can precisely deliver the pouch cell to the disassembly mechanism. Figure 10 shows the detailed layout of the disassembly mechanism, the three stages present in the mechanism, along with the components.



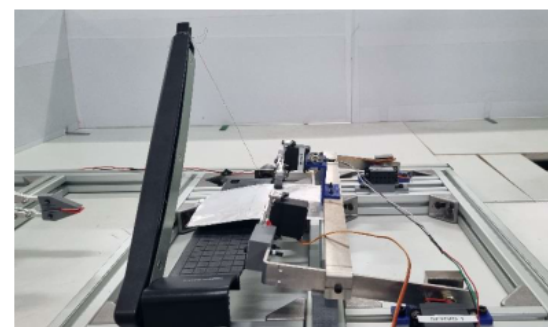
a)



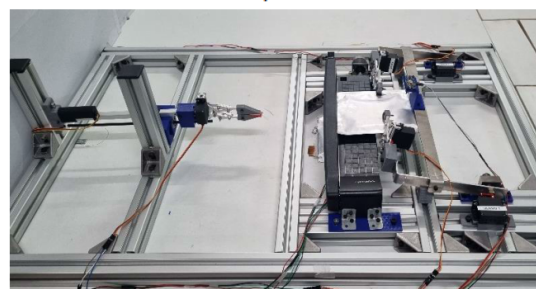
b)



c)



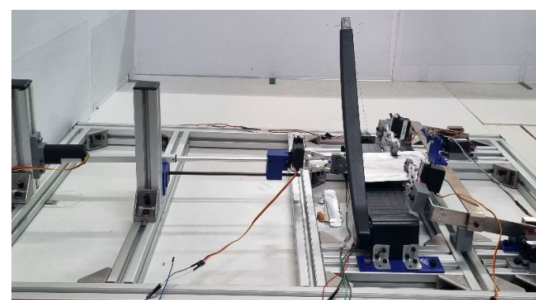
d)



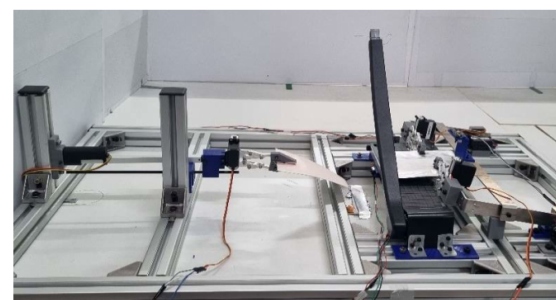
e)



f)



g)



h)

Figure 11. Steps involved in the disassembly process a) The manipulator picks up a pouch cell from the battery b) Placing the pouch cell on stage 1 of the mechanism c) Stage 1 grippers closed d) Pouch cell is placed in position for Stage 2 e) Pouch cell is cut open f) Cutter goes back to open position g) Stage 3 grasps the electrode stack h) Electrode stack extracted

Stage 1 consist of a swivel arm with two grippers, all controlled by servo motors. An ultrasonic sensor is provided on the arm allowing it to detect the presence of the pouch cell, as it is being delivered by the robotic manipulator, to automatically trigger closing of the grippers, as shown in Figure 11 b) and c). Once the grippers are closed, the arm is made to swivel with respect to the base, allowing it to precisely place the pouch cell on the cutting mechanism (Stage 2), as shown in Figure 11 d).

Stage 2 employs an industrial cutter operated by a DC motor with limit switches on both extreme positions. The default position of cutter is open, as shown in Figure 11 d). Once the pouch cell is placed on the cutter the, the motor is actuated to perform a precise cut on the pouch cell, after which the cutter returns to open position, as shown in Figure 11 e), and f).

Stage 3 uses a linear actuator with a gripper at the end to pinch and extract the electrode stack from the pouch cell, as shown in Figure 11 g), and h). The control of the individual stages based on timing and sensor detection (ultrasonic sensor and limit switches) was performed using an Arduino microcontroller. The base frame of the mechanism was built using 40x40 aluminium extrusion.

Multiple trials of battery disassembly were performed, and the average time taken for the complete disassembly operation, starting with the human operator removing the battery cover to the disassembly mechanism extracting the electrode stack, was 4 minutes. It should be noted that the above timing depends on the speed of the manipulator, different stages of the mechanism and the human operator. While this was the best possible time achievable by the proof of concept system, it could be improved significantly through the use of better actuators in the disassembly mechanism, along with a trained human worker.

4.2. Comprehensive assessment of the proposed system

Environmental concerns stemming from the current approach to LIB recycling are evident, with less than 40% of total battery materials being effectively recycled under existing material flow schemes, leading to economic losses and environmental damage due to the unrecovered valuable materials and their subsequent disposal[22]. As of today, the world lacks sufficient recycling infrastructure, with only a few full-scale recycling facilities in operation.

The concept of direct manual recycling in the context of LIBs holds significant promise from both environmental and financial perspectives. It aligns well with the principles of green sustainability and the circular economy, maximizing resource utilization while minimizing the environmental footprint. However, direct manual recycling does face a key challenge - the intricate pretreatment process, which demands substantial manpower and currently confines it mostly to laboratory demonstrations. This labour-intensive aspect renders it economically infeasible for widespread adoption[35].

When we examine the overall recycling costs, which encompass labour, materials, utilities, and general expenses, hydrometallurgy emerges as the most cost-effective process, while pyrometallurgy stands as the most expensive[38]. However, our approach incorporates semi-automated disassembly, which leads to reduced labour expenses, while still maintaining the advantages of direct recycling, making it more economical. Direct, semi-autonomous recycling, allows for maximizing resource use while minimizing environmental impact and keeping labour requirement to minimal.

While the proof-of-concept system presented in this work employs a disassembly mechanism tailor made to suit a specific pouch cells design, it could be extended to accommodate different types of pouch cell variants. This could also be addressed if the battery industry moves towards a standardized cell design.

In summary, the direct recycling of LIBs via a semi-autonomous approach presents a promising solution that combines environmental benefits, such as reduced emissions and energy consumption, with improved financial prospects, including higher revenues compared to traditional pyro- and hydrometallurgical processes. To fully realize its potential,

investments in recycling infrastructure and process optimization are crucial, paving the way for a more sustainable and economically viable approach to LIB recycling.

5. Conclusion

The existing scenario for LIB disposal lacks universal guidelines. Depending on requirements such as the number of batteries exiting the market, current legislation, and available infrastructure, different countries use a variety of procedures, including landfill, incineration, and full or partial recycling. However, the lack of defined processes raises the potential of faulty or irresponsible processing and disposal, resulting in soil, water, and air contamination.

Direct recycling of lithium-ion batteries (LIB) surpasses traditional approaches in terms of efficiency and sustainability. Given this, it becomes important to prioritize the development of a recycling system that considers this direct approach. To this extent, we proposed a Semi-Autonomous robotic system towards efficient recycling of LIBs, showcasing a successful collaboration between human worker, robotic manipulator, and a disassembly mechanism.

The proposed system enables the safe removal of pouch cells from the battery pack and facilitates the subsequent disassembly of the z stack from the pouch cell. The proposed approach could provide significant improvements in the quality of the recycled material as compared to the traditional shredding strategy. Moreover, our approach is adaptable to different battery designs and scalable to handle the increasing volume of LIBs reaching the end-of-life stage.

It should be noted that, detailed experimental validation of the proposed semi-autonomous robotic system over disassembly of multiple battery types and repeated experimental trails is beyond the scope of this concept paper. Instead, the aim of this preliminary study is to offer an efficient and sustainable concept for LIB recycling and underscore the urgency and potential of adopting such innovative recycling strategies to address the pressing challenges posed by LIB disposal.

6. Acknowledgments

This work was supported by Accenture Global Solutions Limited.

7. Patent

The semi-autonomous battery disassembly system presented in this paper is covered by a provisional patent filed with the Indian Patent Office. Patent application number: TEMP/E-1/60202/2023- CHE

References

- Hannappel, Ralf. (2017). The impact of global warming on the automotive industry. AIP Conference Proceedings. 1871. 060001. 10.1063/1.4996530.
- Laura Lander, Tom Cleaver, Mohammad Ali Rajaeifar, Viet Nguyen-Tien, Robert J.R. Elliott, Oliver Heidrich, Emma Kendrick, Jacqueline Sophie Edge, Gregory Offer, Financial viability of electric vehicle lithium-ion battery recycling, *iScience*, Volume 24, Issue 7, 2021, 102787, ISSN 2589-0042,
- Rahman, A., AFROZ, R., SAFRIN, M. (2017). Recycling and Disposal of Lithium Battery: Economic and Environmental Approach. *IIUM Engineering Journal*, 18(2), 238–252.
- International Energy Agency (IEA), *Global EV Outlook 2019: Scaling-up the transition to electric mobility*, OECD Publishing, Paris. <https://doi.org/https://doi.org/10.1787/35fb60bd-en>
- E. Olson, "AprilTag: A robust and flexible visual fiducial system," 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, 2011, pp. 3400-3407
- David Coleman, Ioan A. Șucan, Sachin Chitta, Nikolaus Correll, Reducing the Barrier to Entry of Complex Robotic Software: a MoveIt! Case Study, *Journal of Software Engineering for Robotics*, 5(1):3–16, May 2014.
- Stanford Artificial Intelligence Laboratory et al. (2018). Robotic Operating System. Retrieved from <https://www.ros.org>
- Hyuntae Bae, Youngsik Kim (2021). *Mater. Adv.*, 2021,2, 3234-3250
- Heelan, J.; Gratz, E.; Zheng, Z.; Wang, Q.; Chen, M.; Apelian, D. Current and Prospective Li-Ion Battery Recycling and Recovery Processes. *JOM* 2016, 68, 2632–2638.

10. Georgi-Maschler, T.; Friedrich, B.; Weyhe, R.; Heegn, H.; Rutz, M. Development of a recycling process for Li-ion batteries. *J. Power Sources* 2012, 207, 173–182. 406
11. Wang, X.; Gaustad, G.; Babbitt, C.W.; Richa, K. Economies of scale for future lithium-ion battery recycling infrastructure. *Resour. Conserv. Recycl.* 2014, 83, 53–62. 407
12. Da Costa, A.J.; Matos, J.F.; Bernardes, A.M.; Müller, I.L. Beneficiation of cobalt, copper and aluminum from wasted lithium-ion batteries by mechanical processing. *Int. J. Miner. Process.* 2015, 145, 77–82. 408
13. Espinosa, D.C.R.; Bernardes, A.M.; Tenório, J.A.S. An overview on the current processes for the recycling of batteries. *J. Power Sources* 2004, 135, 311–319. 409
14. Geissdoerfer, M.; Savaget, P.; Bocken, N.M.; Hultink, E.J. The Circular Economy—A new sustainability paradigm? *J. Clean. Prod.* 2017, 143, 757–768. 410
15. Reuter, M.A.; Worrell, E. *Handbook of Recycling*; Elsevier: Amsterdam, The Netherlands, 2014. 411
16. Marshall J, Gastol D, Sommerville R, Middleton B, Goodship V, Kendrick E. Disassembly of Li Ion Cells—Characterization and Safety Considerations of a Recycling Scheme. *Metals*. 2020; 10(6):773. 412
17. N. Wu, Jimmy, Mackenzie, Andrew, Scharma, “Recycling lithium-ion batteries: adding value with multiple lives,” *Green Chem.*, vol. 22, no. 7, pp. 2244–2254, 2020. 413
18. D. J. Garole, R. Hossain, V. J. Garole, V. Sahajwalla, J. Nerkar, and D. Dubal, “Recycle, Recover and Repurpose Strategy of Spent Li-ion Batteries and Catalysts: Current Status and Future Opportunities,” *ChemSusChem*, vol. 13, pp. 1–23, 2020. 414
19. *Energy Environ. Sci.*, 2021, 14, 6099 415
20. Camargos, PH, dos Santos, PHJ, dos Santos, IR, Ribeiro, GS, Caetano, RE. Perspectives on Li-ion battery categories for electric vehicle applications: A review of state of the art. *Int J Energy Res.* 2022; 46(13): 19258–19268. doi:10.1002/er.7993 416
21. S.; Neef, C.; Wicke, T. Trends in Automotive Battery Cell Design: A Statistical Analysis of Empirical Data. *Batteries* 2023, 9, 261. https://doi.org/10.3390/batteries9050261 417
22. Dana Thompson, Charlotte Hyde, Jennifer M. Hartley, Andrew P. Abbott, Paul A. Anderson, Gavin D.J. Harper, To shred or not to shred: A comparative techno-economic assessment of lithium ion battery hydrometallurgical recycling retaining value and improving circularity in LIB supply chains, *Resources, Conservation and Recycling*, Volume 175, 2021, 105741, ISSN 0921-3449, 418
23. Skeete, J.P., Wells, P., Dong, X., Heidrich, O., Harper, G., 2020. Beyond the Event horizon: Battery waste, recycling, and sustainability in the United Kingdom electric vehicle transition. *Energy Research and Social Science* 69, 101581. 419
24. Harper, G., Sommerville, R., Kendrick, E. et al. Recycling lithium-ion batteries from electric vehicles. *Nature* 575, 75–86 (2019). 420
25. *Renewables* 2023, 1, 294–315 421
26. Wegener, K., Chen, W. H., Dietrich, F., Dröder, K., Kara, S. Robot assisted disassembly for the recycling of electric vehicle batteries. *Proc. CIRP* 29, 716–721 (2015). 422
27. Dornfeld, D. A. & Linke, B. S. (eds) *Leveraging Technology for a Sustainable World*. (Proc.19th CIRP Conf. on Life Cycle Engineering) (Springer, 2012). 423
28. Arora, S. & Kapoor, A. in *Behaviour of Lithium-Ion Batteries in Electric Vehicles* (eds Pistoia, G. & Liaw, B.) 175–200 (Springer, 2018). 424
29. Pehlken A, Albach S, Vogt T. Is there a resource constraint related to lithium ion batteries in cars? *Int J Life Cycle Assess.* 2017; 22(1): 40–53. 425
30. Wegener K, Andrew S, Raatz A, Dröder K, Herrmann C. Disassembly of electric vehicle batteries using the example of the Audi Q5 hybrid system. *Proc CIRP* 2014; 23: 155–160 426
31. Harper G, Sommerville R, Kendrick E, et al. Recycling Li-Ion batteries from electric vehicles. *Nature*. 2019; 575: 75–86. 427
32. Tan, WJ, Chin, CMM, Garg, A, Gao, L. A hybrid disassembly framework for disassembly of electric vehicle batteries. *Int J Energy Res.* 2021; 45: 8073–8082. 428
33. Kathrin Wegener, Wei Hua Chen, Franz Dietrich, Klaus Dröder, Sami Kara, Robot Assisted Disassembly for the Recycling of Electric Vehicle Batteries, *Procedia CIRP*, Volume 29, 2015, Pages 716–721, ISSN 2212-8271, 429
34. E. Olson, “AprilTag: A robust and flexible visual fiducial system,” 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, 2011, pp. 3400–3407, doi: 10.1109/ICRA.2011.5979561. 430
35. Wang, X-T, Gu, Z-Y, Ang, EH, Zhao, X-X, Wu, X-L, Liu, Y. Prospects for managing end-of-life lithium-ion batteries: present and future. *Interdiscip Mater.* 2022; 1: 417–433. doi:10.1002/idm2.12041 431
36. J. B. Dunn, L. Gaines, J. C. Kelly, C. James and K. G. Gallagher, *Energy Environ. Sci.*, 2015, 8, 158–168. 432
37. Pražanová, A.; Knap, V.; Stroe, D.-I. Literature Review, Recycling of Lithium-Ion Batteries from Electric Vehicles, Part II: Environmental and Economic Perspective. *Energies* 2022, 15, 7356. 433
- en15197356 434
38. Laura Lander, Tom Cleaver, Mohammad Ali Rajaeifar, Viet Nguyen-Tien, Robert J.R. Elliott, Oliver Heidrich, Emma Kendrick, Jacqueline Sophie Edge, Gregory Offer, Financial viability of electric vehicle lithium-ion battery recycling, *iScience*, Volume 24, Issue 7, 2021, 102787, ISSN 2589-0042, 435

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content. 461